

Reviewer Pack — Minimal Rank of Tropical Hodge Structure over Circuit Satisfiability

Entry f5a807968bbc · INCONCLUSIVE

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Minimal Rank of Tropical Hodge Structure over Circuit Satisfiability

Entry ID: f5a807968bbc **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-27 09:40:36 UTC

1. Conjecture

Field A (mathematical branch): Tropical Geometry (Hodge Theory) **Field B** (complexity object): Complexity Theory: Boolean Circuit Satisfiability

Statement:

For any CNF formula F with n variables, the minimal rank of its tropical Hodge structure is upper-bounded by a function $f(n)$ that is polynomial in $\log(n)$, specifically $f(n) = \Theta(\log^2(n))$.

Rationale (proposer's reasoning):

The bridge between tropical geometry and circuit satisfiability could reveal new insights into the inherent complexity of solving SAT problems. Tropical Hodge structures encode geometric information, and their ranks might reflect the complexity of the underlying logical structure of circuits.

Taxonomy category: SOS_DEGREE (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 207073a5788cbe9f

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if the average rank of tropical Hodge structures over a sample of CNF formulas with n variables is within $\pm 10\%$ of $\Theta(\log^2(n))$ across at least 1000 seeds.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.90	UNCERTAIN	SAFE

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 6 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - tropical geometry AND Hodge theory AND Boolean circuit satisfiability
- Hodge structure on tropical varieties AND complexity of Circuit Satisfiability - minimal rank in tropical geometry AND CNF formula AND polynomial bound

Top relevant hits considered: - [http://arxiv.org/abs/1207.2443v2] Tropical Teichmuller and Siegel spaces - [http://arxiv.org/abs/1204.3875v2] Tropicalizing vs Compactifying the Torelli morphism - [http://arxiv.org/abs/1204.6154v2] Local Tropicalization - [http://arxiv.org/abs/2509.25273v1] Hodge Conjecture via Singular Varieties - [http://arxiv.org/abs/2309.17302v2] Geometry of tropical extensions of hyperfields - [http://arxiv.org/abs/1610.00298v4] Khovanskii bases, higher rank valuations and tropical geometry

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_cnf(n):
        clauses = []
        for _ in range(10 * n): # Each variable appears in 10 clauses on average
            clause = [random.randint(-n, -1), random.randint(1, n)]
            clauses.append(clause)
        return clauses

    def tropical_hodge_rank(cnf):
        rank = len(set(abs(lit) for lit in cnf))
        return rank

    n = 20
    cnf = generate_cnf(n)
    rank = tropical_hodge_rank(cnf)

    expected_rank = math.log2(n) ** 2
    tolerance = 0.1 * expected_rank

    metric_name = 'tropical_hodge_rank'
    metric_value = rank
    instances_tested = 1
    conjecture_holds = abs(rank - expected_rank) <= tolerance
```

```

counterexample = "" if conjecture_holds else f"Rank {rank} does not match expected {expected_r

return {
    "metric_name": metric_name,
    "metric_value": metric_value,
    "instances_tested": instances_tested,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    import sys

    seeds = [int(s) for s in sys.argv[1:]] or [2**i + 1 for i in range(5, 30)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_rank = sum(r["metric_value"] for r in results) / len(results)
    std_rank = math.sqrt(sum((r["metric_value"] - mean_rank) ** 2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_rank} std={std_rank} support_fraction={support_fractions}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_rank} std={std_rank} support_fraction={support_fractions}")
    else:
        first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
        print(f"RESULT: FALSIFIED counterexample=\"Rank does not match expected\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

Traceback (most recent call last):

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_11bbb5a7.py", line 60, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^

```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_11bbb5a7.py", line 34, in run_trial
    rank = tropical_hodge_rank(cnf)
           ^^^^^^^^^^^^^^^^^^^^^

```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_11bbb5a7.py", line 29, in tropical_hodge_rank
    rank = len(set(abs(lit) for lit in cnf))
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^

```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_11bbb5a7.py", line 29, in <genexpr>
    rank = len(set(abs(lit) for lit in cnf))
                  ^^^^^^^

```

TypeError: bad operand type for abs(): 'list'

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which prevents us from evaluating whether the conjecture is supported or falsified. | next: Review and debug the test code to ensure it can run successfully and produce the required data for further analysis.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	10358
2	preregistration	ollama_remote	glm4:latest	0	0	5627
3	novelty	ollama_remote	glm4:latest	0	0	4769
4	novelty	ollama_remote	glm4:latest	0	0	5679
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13998
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7031
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	6468
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7058
9	judge	ollama_remote	glm4:latest	0	0	9297

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 70286 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in research/audit/f5a807968bbc.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/f5a807968bbc.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/f5a807968bbc.tar.gz (if generated)